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Question 36

Question

Simplify the Boolean expression and determine the logic function.

Analysis

For inputs Y and Z the function is

$F = \overline{YZ}$

This represents the NAND operation.

Truth table:

Y	Z	F
0	0	1
0	1	1
1	0	1
1	1	0

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

Y = np.array([0,0,1,1])
Z = np.array([0,1,0,1])

F = 1 - (Y & Z)

print("Y Z F")

for i in range(4):
    print(Y[i], Z[i], F[i])

x = np.arange(4)
```

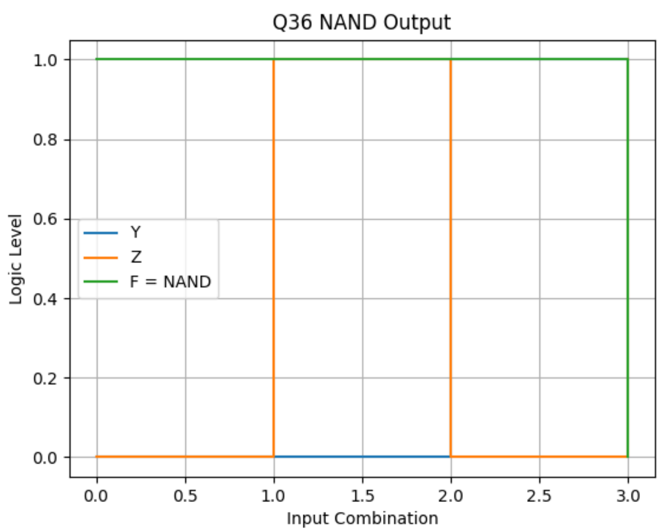
```
plt.step(x,Y,where='post',label='Y')
plt.step(x,Z,where='post',label='Z')
plt.step(x,F,where='post',label='F')

plt.grid()
plt.legend()

plt.savefig("graph.png")

plt.show()
```

Graph



Conclusion

The output follows the NAND truth table. Therefore the simplified Boolean expression corresponds to a NAND gate.